

PnRInfo : Interactive Tactical Information Visualization for Pick and Roll Event

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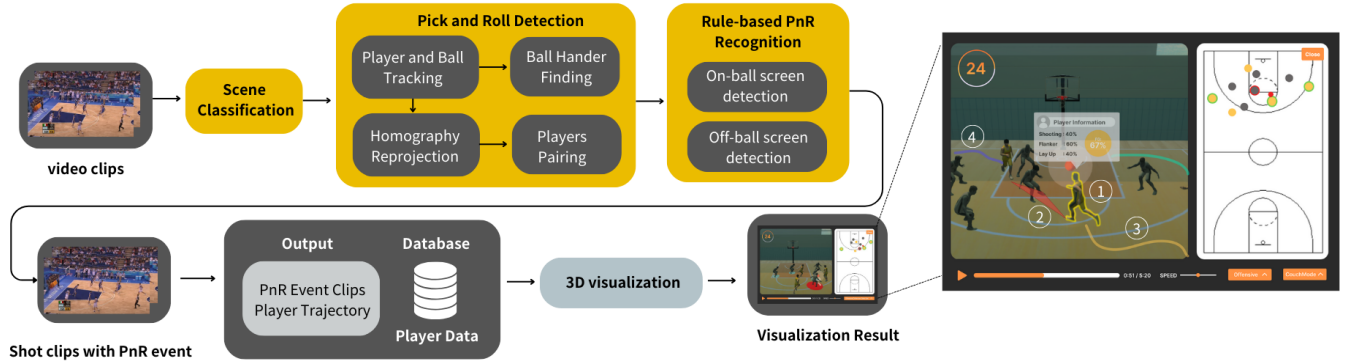


Figure 1: Pipeline of the proposed PnRInfo visualization system. In the right figure, 1 demonstrates the visualization of the Important Player Highlight, 2 displays the Effective Attacking Space, while 3 and 4 show the trajectories of different players.

Abstract

With pick-and-roll (PnR) being a commonly used tactic in basketball, providing a thorough analysis of it proves to be beneficial to improving a basketball team's overall performance. In this work, we propose PnRInfo, a tool for automatically detecting PnR plays within basketball footage and visualizing respective tactical information in a 3D simulation space to facilitate interactive discussion of PnR tactics.

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1 Introduction

Analyzing game footage is crucial for coaches aiming to improve their team's level of strategy, especially in fast-paced sports like basketball. As manual analysis is time-consuming and labor-intensive, automation through computer vision technology is an efficient and cost-effective alternative. However, clear and effective visualization is key to conveying insights from this data. Several tools exist that target the visualization of basketball game data. For example, Hoop-InSight [Fu and Stasko 2024] is an interactive visualization system

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for shot analysis, while BkViz [Losada et al. 2016] focuses on how players perform through interactive visuals. While these works focus on broader aspects of basketball games, such as the overall shooting performance or player trajectories, there remains a lack of tools made suitable for training tactics. Preliminary interviews with professional coaches highlight that pick-and-roll (PnR) is a key tactic, essential for creating scoring opportunities and improving a team's chances of winning. PnR involves a player blocking a defender to create space for a teammate. We define PnR into two distinct categories: on-ball PnR, involving the ball-handler, and off-ball PnR, between two non-ball-handling players.

In this work, we propose PnRInfo, a framework for automatically detecting PnR plays within a basketball game and visualizing tactical information in 3D to facilitate an interactive discussion and understanding of each play. Our contributions are two-fold: (1) We propose an efficient method to detect PnR events by pairing offensive and defensive players using bipartite matching and the Hungarian algorithm, followed by a rule-based approach to identify on-ball and off-ball PnRs. (2) We design a novel 3D visualization tool for PnR tactics analysis, allowing for easier observation and explanation of player movements from various perspectives. It maps player paths and provides informative data to help users understand the best actions for each situation. We focus on the effectiveness of PnR tactics for offensive players and develop interactive tools that allow coaches and players to better communicate tactical concepts.

2 Method

Fig. 1 shows the PnRInfo system framework with three modules: scene classification (using ResNet50), PnR detection (on-ball/off-ball), and 3D tactical visualization for displaying information.

Pick and Roll Detection: For each court view play, we employ ByteTrack [Zhang et al. 2022] for player tracking, WASB-SBDT [Tarashima et al. 2023] for ball tracking, and OmniScorer [Cheng et al. 2023] for homography estimation. The lower midpoint of each player's bounding box is projected onto a bird's-eye view using the homography matrix. Based on the definition of pick-and-roll (PnR), we implement a player-pairing method to identify which defensive player guards which offensive player. A rule-based method then detects on-ball and off-ball PnR by analyzing how screeners position themselves near defenders. We model this as a bipartite graph, with edges weighted by distance, and use the Hungarian algorithm to solve the minimum bipartite matching problem. Additionally, by analyzing bounding box overlaps with the basketball, we identify the ball handler within each time window. A set of carefully designed rules is then utilized to recognize whether the play includes an on-ball/off-ball PnR event and to identify the players involved in the PnR. We detect on-ball screens based on player and ball positions, referring to the definition that "instances where an offensive player, who is not in possession of the ball, gets into a close distance to the defender of the ball handler." Off-ball PnR can be detected directly based on the distances between players. For each off-ball offensive player, we ascertain their defender through the previously described player pairing process. Then, we prematurely identify players involved in off-ball PnR actions by examining whether another offensive player is approaching the paired defender. To

avoid false positives, we confirm that the screener's feet remains stationary during PnR in accordance with regulation.

Data Visualization and Interaction: We interviewed four professional basketball coaches to determine key insights for analyzing PnR tactics and designed the system to offer both teams' perspectives. In offensive mode, defenders are grayed out to highlight the offense's positioning and movement, and vice versa for defensive mode. We use embedded visuals to display PnR tactics on-screen, aiding coaches in analyzing footage more efficiently. The information displayed includes *Important Players Highlight*, *Player Trajectories*, *Player stats*, *Effective Attacking Space Visualization*. This visualization uses the distance between the ball-handler's defender and the passing trajectory, with halos for key players and colors for their paths, aiding coaches in tracking movements. Additionally, we use arcs around the defenders' feet relative to their offensive players to display defensive effectiveness. *Effective Attacking Space Visualization* shows the quality of attacking opportunities created by PnR, using the triangular area between an opposing pair of players and a screener. A larger green area signifies a higher quality PnR, while a smaller red area indicates limited opportunities. In the 3D environment, users can select a 2D view resembling tactic boards for basic strategies. The Notation Interface lets coaches annotate in "Coach Mode" and replay in "Game Mode."

3 Results

We utilized the MultiSports dataset [Li et al. 2021], which contains annotated event intervals with the players' bounding boxes, to demonstrate the accuracy of our PnR detection module. The accuracy of the predicted time intervals of PnR occurrences is calculated based on the Precision, Recall, and F1-score with different tIoU thresholds. With an tIoU threshold of 0.3, the on-ball PnR and off-ball PnR detection performance achieved F1-scores of 0.70 and 0.45, respectively.

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